

Code Review — RPG Combat Kata

Repository: rpg-combat · **Branch:** `main` @ `2fbdb7c` · **Date:** 2026-06-25 **Baseline health:** 33/33 vitest tests passing · eslint clean (functions ≤ 10 lines, file ≤ 150 , complexity ≤ 5) **Method:** per-file architectural fan-out + a 2-skeptic adversarial panel.

Status: Kata complete — ending here by choice

This project is a **coding kata**. The implementation fully satisfies every acceptance criterion in [docs/backlog/](#) (Stories 1–5), built test-first with one test per acceptance criterion and characterization tests mutation-verified.

The findings below were produced by a final code review for completeness. **We are consciously choosing to end the kata here and not remediate them.** They are recorded as documentation of known trade-offs, not as a backlog of pending work. Most are hardening concerns (input validation, depletion semantics) or design refinements that fall outside the kata's stated acceptance criteria; a couple are defensible design choices the review panel ultimately rejected as non-bugs.

High — correctness gaps

H1 · A `HealingObject` never depletes

`HealingObject.drawHealing()` returns `this.health` but never debits the object, so a single healing object can heal infinitely. AC 04 says a character gains health "up to the lesser of the object's **remaining** amount and the Character's own max" — *remaining* implies depletion. No test reads object health after a draw, so it is invisible to the suite. *Suggested fix:* grant `min(remaining, character.headroom)`, debit the object by the granted amount, and add a depletion/second-draw test.

H2 · `Character.takeDamage` is public, bypassing every guard

`takeDamage` is the seam weapons and magical objects rely on, but being public it bypasses the self-check, ally-check, and level modifiers in `dealDamage`. It is the mechanism behind H3 and M1. *Suggested fix:* route weapon/object damage through a narrower internal path and tighten visibility, or document this as the kata's deliberate seam.

H3 · Negative damage heals and corrupts leveling

`takeDamage(-500)` increases health (no floor on the heal side) and drives the level-up `damageBuffer` negative, stalling future level-ups. The guarded path is affected too: `effectiveDamage` only multiplies, never floors, so a negative `dealDamage` amount can heal an enemy. No acceptance criterion permits negative damage. *Suggested fix:* validate `amount >= 0` at the `dealDamage` / `takeDamage` boundary.

Medium

M1 · Weapon attacks bypass self/ally guards

`attackWith` delegates straight to `weapon.attack(target)` → `target.takeDamage(...)`, routing around the self- and ally-guards. A character can weapon-attack an ally or itself. Story 4 "depends on Story 3 for ally/non-ally targeting rules," so intent is debatable — and it is untested either way. (Fixed damage bypassing *level modifiers* is correct per AC.)

M2 · `startingLevel` is unvalidated

`new Character(0)` yields a survival threshold of $0 \times 1000 = 0$, so the first non-lethal hit grants a level. `new Character(11)` exceeds the level-10 cap at construction (the cap is enforced only in `gainLevel`). *Suggested fix*: clamp/validate `startingLevel` to `[1, 10]`.

M3 · A dead character can heal an ally

`healCharacter` checks `!target.isAlive`, but a dead `this` healing a living ally passes. Only dead-self-heal is tested. *Suggested fix*: also guard `!this.isAlive`.

M4 · A destroyed (0-durability) weapon can still attack

No `isDestroyed` guard before attacking; a spent weapon deals full damage. Inferred, not a spec violation, and untested.

Low / Nits

- **L1** · `instanceof Character` in `heal` — a runtime type-test against the `Target` marker interface; cleaner as a polymorphic `receiveHealing` on `Target`.
 - **L2** · `Target` is a degenerate marker interface — only `health` + an always-false `belongsTo`; the `instanceof` re-narrowing defeats its polymorphic purpose.
 - **L3** · **Duplicated health mutation** — `currentHealth + takeDamage + Math.max(0, ...)` appear in both `Character` and `MagicalObject`; a shared `Health` value object would remove it.
 - **L4** · `drawFrom` reads the source before the dead-check — harmless ordering smell.
 - **L5** · **Fractional health** — odd damage $\times 0.5$ can yield `.5` health; no AC requires integers.
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Test-coverage holes worth noting

- No depletion / second-draw test for a `HealingObject` (would expose H1).
- `drawFrom` never exercises the character-max clamp — deleting the clamp leaves all tests green.
- Ally-heal caps at the *ally*'s max, not the healer's — untested (a swap to `this.maxHealth`

would pass everything).

- The "unmodified damage" test uses level difference 2, never the exact **4** boundary.
 - Ally guards are asserted in only one direction (never the reverse).
 - No level-10 cap test via the survived-damage path (only the faction path is tested).
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Reviewed and rejected as non-bugs

- **"Level-up discards overflow damage"** — resetting `damageBuffer` to 0 on level-up is a defensible reading of AC 05 ("cumulative ... across any number of battles" toward one threshold); the spec only describes single-level transitions. Not a defect.
 - **"One hit grants at most one level"** — no AC describes multi-threshold single hits; the level-10 cap bounds the blast radius. Non-finding.
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What's solid

- **Damage modifiers** — all three branches (halve at 5+ above, boost at 5+ below, unmodified within 5) are correct and tested.
- **Faction leveling** — `factionsEverJoined` is a separate Set that never shrinks on `leave`, so "a gained level is never lost" holds; rejoining is idempotent; 3→L2 and 6→L3 verified.
- **Level-10 cap** — a single choke point in `gainLevel`, guarded `< MAX_LEVEL`.
- **Lethal-blow-grants-no-level** — the `if (this.isAlive)` gate correctly defers leveling past damage resolution, tested.
- **Destruction semantics** — `isDestroyed (=== 0)` is consistent with the `Math.max(0, ...)` floor.
- **Capability rejection** — heal-a-weapon and attack-with-a-healer reject at runtime via base-class throws, as the acceptance criteria require; tested.